International Journal of
Radiation Oncology
biology • physics

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Clinical Investigation: Central Nervous System Tumor

Radiogenic Side Effects After Hypofractionated Stereotactic Photon Radiotherapy of Choroidal Melanoma in 212 Patients Treated Between 1997 and 2007

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Received Nov 8, 2010, and in revised form Jun 1, 2011. Accepted for publication Jun 3, 2011

Summary

This article describes secondary radiogenic side effects occurring after hypofractionated stereotactic photon radiotherapy of choroidal melanoma with a median follow-up of over 5 years. The five most common late effects were radiation retinopathy, optic neuropathy, cataract development, neovascular glaucoma and corneal epithelium defects. Hypofractionated stereotactic photon radiotherapy thus shows a low to moderate rate of adverse late effects comparable to those seen after proton beam radiotherapy.

Purpose: To evaluate side effects of hypofractionated stereotactic photon radiotherapy for patients with choroidal melanoma.

Patients and Methods: Two hundred and twelve patients with choroidal melanoma unsuitable for ruthenium-106 brachytherapy or local resection were treated stereotactically at the Medical University of Vienna between 1997 and 2007 with a Linac with 6-MV photon beams in five fractions with 10, 12, or 14 Gy per fraction. Examinations for radiogenic side effects were performed at baseline and every 3 months in the first 2 years, then every 6 months until 5 years and then once a year thereafter until 10 years after radiotherapy. Adverse side effects were assessed using slit-lamp examination, funduscopy, gonioscopy, tonometry, and, if necessary, fundus photography and fluorescein angiography. Evaluations of incidence of side effects are based on an actuarial analysis.

Results: One hundred and eighty-nine (89.2%) and 168 (79.2%) of the tumors were within 3 mm of the macula and the optic disc, respectively. The five most common radiotherapy side effects were retinopathy and optic neuropathy (114 cases and 107 cases, respectively), cataract development (87 cases), neovascular glaucoma (46 cases), and corneal epithelium defects (41 cases). In total, 33.6%, 38.5%, 51.2%, 75.5%, and 77.6% of the patients were free of any radiation retinopathy, optic neuropathy, cataract, neovascular glaucoma, or corneal epithelium defects 5 years after radiotherapy, respectively.

Conclusion: In centrally located choroidal melanoma hypofractionated stereotactic photon radiotherapy shows a low to moderate rate of adverse long-term side effects comparable with those after proton beam radiotherapy. Future fractionation schemes should seek to further reduce adverse side effects rate while maintaining excellent local tumor control. © 2012 Elsevier Inc.

Keywords: Choroidal melanoma, Hypofractionated stereotactic photon radiotherapy, Adverse side effects, Juxtapapillary tumors

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Conflicts of interest: none.

Int J Radiation Oncol Biol Phys, Vol. 83, No. 1, pp. 121−128, 2012
0360-3016/$ - see front matter
doi:10.1016/j.ijrobp.2011.06.1957